Mapping Legislative Attention to the Social Determinants of Mental Health: A Text Analysis of State Bills, 2021-2024

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Abstract

Write your abstract here following the recommendations for an APA empirical abstract. Must not exceed 150 words, providing a concise summary of your research question, methodology, key findings, and implications. The abstract will appear in italic text at the top of the first column.

Keywords: keyword1, keyword2, keyword3, keyword4, keyword5

1. Introduction

The social determinants of health (SDOH) represent the fundamental environmental, social, and economic conditions in which individuals are born, grow, live, work, and age that shape their overall well-being and health outcomes (Kirby & Slone, n.d.; *Social Determinants of Health - Healthy People 2030 | Odphp.health.gov*, n.d.). While these determinants influence both physical and psychological health, the core domains particularly relevant to mental health and this study include economic stability, education access and quality, health care access and quality, neighborhood and built environment, social and community context, and discrimination, equity, and rights (Kirby & Slone, n.d.; *Social Determinants of Health - Healthy People 2030 | Odphp.health.gov*, n.d.). Crucially, experiences of disadvantage follow a distinct social gradient, meaning that individuals occupying lower social positions face compounding mental health risks that accumulate over the life course through persistent exposure to chronic stress and structural inequities (Alegría et al., 2018). Furthermore, this relationship is highly reciprocal, as untreated mental health disorders can actively precipitate further social disadvantage–such as unemployment, marital instability, or housing loss–creating a harmful feedback loop that deeply entrenches inequality (Alegría et al., 2018; Rotter et al., 2022).

These foundational factors function as upstream drivers, acting as the fundamental causes that shape downstream living conditions and immediate health risks (Alegría et al., 2018; Erickson et al., 2023). Consequently, effectively addressing mental health requires a critical shift away from focusing exclusively on individual pathology and biomedical models of disease, moving instead toward an examination of the systemic inequities and structural conditions that generate psychological distress (Kirby & Slone, n.d.; Macintyre et al., 2018). While individual-level clinical interventions are important, public policy serves as an important mechanism for shaping these overarching environmental and socioeconomic conditions (Gostin et al., 2019; Kirby & Slone, n.d.). By establishing rules, structuring societal systems, and directing the equitable distribution of resources, the law can directly mediate these upstream social determinants, providing an important tool to restructure societal frameworks and facilitate a healthier, more equitable society (Erickson et al., 2023; Gostin et al., 2019; Kirby & Slone, n.d.).

Despite widespread recognition of the critical importance of the SDOH, existing scholarship does not systematically map how state mental health bills address these specific domains across varying legislative and political contexts. This gap persists largely because the current literature is dominated by qualitative examples and narrow case studies that, while providing deep insights, cannot easily capture broad structural trends. These legal texts create a methodological challenge because they consist of large volumes of unstructured language that are difficult to analyze systematically through manual review (Grimmer & Stewart, 2013; Sewerin et al., 2023).

To overcome the costs and logistical barriers of manual coding, text-as-data methods provide a scalable way to analyze large collections of political texts (Grimmer & Stewart, 2013). By leveraging these computational techniques, researchers can extract meaningful patterns from large-scale legislative corpora that were previously difficult to process systematically at scale (Grimmer & Stewart, 2013). In this study, legislative attention refers to the extent and pattern of social determinant language observed in state mental health bill texts. To address the current gap in the literature, this study maps legislative attention to the social determinants of mental health in a LegiScan-derived, keyword-filtered cor-

pus of introduced state mental health bills from ten states between 2021 and 2024, comparing patterns across domains, states, session years, and party control.

2. Literature Review

Mental health is shaped not only by clinical care and individual risk factors, but also by upstream social, economic, and environmental conditions. While downstream factors typically involve individual-level risks, health behaviors, and direct clinical interventions, upstream determinants represent the fundamental societal, political, and economic structures that act as the root causes of these health inequities (Chibani, 2025; Whitman et al., n.d.). A substantial body of evidence demonstrates that experiences of socioeconomic disadvantage–including poverty, unemployment, housing instability, social exclusion, and unsafe physical environments–are consistently linked to worse mental health outcomes and higher rates of psychiatric illness (Kirby & Slone, n.d.; Macintyre et al., 2018). Because these foundational determinants influence an individual's health and well-being through diverse, interacting pathways across the lifespan, utilizing a domain-based lens that categorizes these factors into distinct areas like economic stability, neighborhood and built environment, and social and community context provides a highly useful framework for understanding and addressing them (*Social Determinants of Health - Healthy People 2030 | Odphp.health.gov*, n.d.; Storm et al., 2025).

A domain-based framework is useful because the social-determinants literature groups upstream influences into recurring categories rather than treating them as one undifferentiated factor. For instance, the Healthy People 2030 initiative organizes these conditions into five distinct domains: Economic Stability, Education Access and Quality, Health Care Access and Quality, Neighborhood and Built Environment, and Social and Community Context (*Social Determinants of Health - Healthy People 2030 | Odphp.health.gov*, n.d.). These broad categories naturally map onto themes already visible in the mental health literature, where researchers consistently link psychiatric outcomes to specific upstream factors like income and job insecurity, poor education, limited care access, housing instability, and lack of social support or exposure to violence (Kirby & Slone, n.d.; *Leveraging Data on the Social Determinants of Health*, n.d.). Furthermore, systemic discrimination, health equity, and human rights deserve explicit attention in mental-health-focused studies because experiences of social exclusion, structural racism, and stigma act as fundamental causes of psychological distress that compound vulnerabilities across all other domains (Kirby & Slone, n.d.; Macintyre et al., 2018; Whitman et al., n.d.).

If social determinants shape mental health, then law and public policy are important mechanisms through which governments structure those conditions. Through statutory and regulatory frameworks, the law directly affects the operations of public and private institutions, dictates the equitable or inequitable distribution of resources, and shapes both eligibility for and the delivery of health services (Gostin et al., 2019). Because of its pervasive influence, policy possesses a dual potential: it can either actively reduce harm by building strong health systems, or it can reproduce social exclusion and entrench stigma (Gostin et al., 2019). Poorly designed or punitive laws can institutionalize inequality and create systemic barriers that discourage marginalized populations from seeking treatment (Gostin et al., 2019). Consequently, legislative language matters immensely because policy texts do more than merely record objective administrative actions; the specific wording used actively frames societal problems, conceptualizes target populations, and signals governmental priorities and approaches to intervention (Popa, 2025).

Existing state-level mental health scholarship suggests that attention to social determinants is real but uneven across domains and jurisdictions. As Rotter and colleagues demonstrate, state mental health authorities display broad recognition and concern regarding the relevance of social determinants, yet this awareness suffers from a lack of standardized approaches and limited intentional implementation of specific screening tools or intervention mandates (Rotter et al., 2022). Moreover, programmatic focus is highly skewed; some determinants–especially housing instability and health care access and quality–receive significantly more attention and targeted action than other crucial factors such as education, food insecurity, and discrimination, equity, and rights (Rotter et al., 2022). Consequently, this variation across states is substantively meaningful and not merely a data convenience, as these diverging regional priorities create profound inequities and dictate fundamentally different lived experiences within the mental health system depending on where an individual resides (Nelson et al., 2022; Rosenberg et al., 2023).

Although prior scholarship shows that social determinants matter for mental health and that law helps structure those conditions, less is known about how state mental health bills systematically express these concerns across domains and contexts. While the existing literature thoroughly explains why SDOH matter for overall well-being and why public policy is a crucial mechanism for addressing structural health inequities (Kirby & Slone, n.d.), current research does not fully map domain-specific language across a comprehensive, multi-stage bill corpus (Nelson et al., 2022; Sewerin et al., 2023). Furthermore, variation across states, session years, and party control–such as shifts in interparty compe-

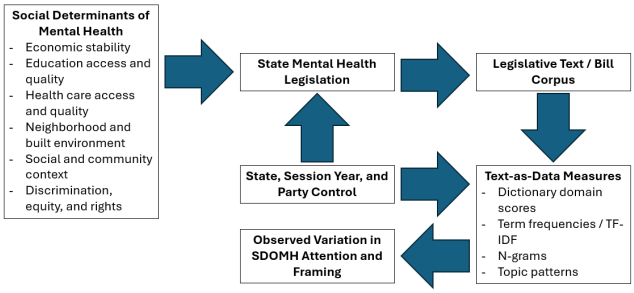

Figure 1: This framework depicts how social determinants of mental health are translated into observable patterns of legislative attention through state mental health bills, with text-derived measures used to compare variation across states, session years, and party control.

tition, electoral turnover, and differing regional governance–are highly plausible and important factors in shaping public policy (TAKEDA, 1987), but they are still underexamined in this specific legislative-text context (Popa, 2025). To address this critical gap, this study contributes to the field by mapping social determinant language in state mental health bills across domains, states, session years, and party control.

3. Conceptual Framework

Because law can greatly affect the social determinants of mental health, state policy is an important site for understanding how governments respond to those conditions. This project treats state mental health bills as institutional signals of which social determinants receive legislative attention. Bill text serves as the observable link between scholarship on social determinants and measurable patterns of legislative attention. The analysis compares these patterns across states, session years, and party control, with the goal of mapping policy attention rather than estimating downstream mental health outcomes.

4. Research Questions

RQ1. How frequently do state mental health bills mention specific social determinant domains, and which domains are most prominent? RQ1a. Which domain terms are most frequent by TF? RQ1b. Which terms are most distinctive by TF-IDF? RQ1c. Which bigrams and trigrams recur most often within each domain?

RQ2. How does the use of social determinant domain language vary by state, session year, and party control? RQ2a. Which states use the highest share of neighborhood and built environment or discrimination, equity, and rights language? RQ2b. Does domain language increase or decrease over time? RQ2c. Are certain domains more common under Democratic, Republican, or divided control?

RQ3. What terms, phrases, and named entities distinguish bills with high versus low levels of social determinant language? RQ3a. Which words and phrases are most distinctive in high-domain bills? RQ3b. Which organizations, agencies, programs, or place names appear most often in high-domain bills? RQ3c. Do high-domain bills reference schools, housing authorities, Medicaid, or local governments more often than low-domain bills?

RQ4. What latent themes emerge in mental health bills, and how do those themes relate to concrete social determinant domains? RQ4a. Which topics are most common in the corpus? RQ4b. Which topics co-occur with high neighborhood and built environment, education access and quality, or discrimination, equity, and rights scores? RQ4c. Do some states or session years show stronger topic-domain combinations than others?

5. Methodology

5.1 Data Collection and Corpus Construction

This study analyzes introduced state legislative bills from ten U.S. states (New York, Illinois, Texas, Florida, California, Pennsylvania, Washington, Georgia, Wisconsin, and Ohio) across regular sessions from 2021–2024. Bills and metadata were drawn from LegiScan's public historical API archives, and the corpus was restricted to introduced versions to ensure comparability at a common legislative stage. Mentalhealthrelevant bills were identified by applying a fixed keyword screen (e.g., "mental health," "behavioral health," "suicide," "civil commitment," "substance use disorder") to bill titles and descriptions, reducing the candidate pool to 3,395 bills. For these bills, full text was retrieved from LegiScan, converted to UTF-8 plain text using format-appropriate tools, and validated through automated checks and manual spotreviews. After excluding special sessions and nonbill records, the final analytic corpus consists of 3,169 introduced bills paired with metadata. A key limitation is that PDFtotext conversion preserves lexical content but not visual markup (e.g., strikethroughs), so repealed and inserted language may be treated identically in downstream tokenbased analyses.

5.2 Data Preparation

Two contextual variables—session year and party control—were prepared for subgroup comparisons in RQ2

and descriptive stratification throughout the analysis. Session year was coded as the start year of the regular legislative session in which the bill was introduced (2021 or 2023), and party control at session opening was classified as Democratic trifecta, Republican trifecta, or divided government using National Conference of State Legislatures and Ballotpedia data. These variables are used as descriptive comparison dimensions rather than as causal treatments.

From the master corpus table, the analysis constructs a set of derived tables used across the research questions: a token table for unigrambased frequency and TF–IDF analysis, an ngram table for bigram and trigram analysis, a namedentity table for organization and place extraction, a domaindictionary table for Social Determinants of Health (SDOH) measurement, and a topic table of document–topic proportions for RQ4. All text processing was implemented in R using a tidytext workflow for tokenization and documentterm representations, with spacyr/spaCy used for named entity recognition. Preprocessing decisions were tailored to preserve interpretability—using raw, unstemmed tokens for dictionary scoring, ngrams, and named entities, and lemmatized tokens for TF–IDF and topic modeling—and to reduce noise by removing standard stopwords, pure numbers, very short tokens, and highfrequency legislative boilerplate. Very long bills were truncated for NER only to stay within spaCy's length constraints, and the resulting entities were manually inspected to remove generic procedural artifacts and to restrict types to substantively interpretable organizations and geopolitical entities.

5.3 Dictionary and Measures

To operationalize "legislative attention" to upstream determinants, this study uses a custom domain dictionary that maps bill language to six socialdeterminant domains relevant to mental health: Economic Stability; Education Access and Quality; Health Care Access and Quality; Neighborhood and Built Environment; Social and Community Context; and Discrimination, Equity, and Rights. Dictionary methods are appropriate here because they provide a transparent, deductive strategy in which concept definitions are specified beforehand and applied consistently across a large corpus.

Dictionary development combined deductive term selection from the socialdeterminants literature and the Healthy People 2030 SDOH framework with an inductive validation pass on a sample of bills to add domainrelevant language and remove ambiguous or overly generic terms. Candidate terms were retained if they clearly referred to social conditions or structural factors, and if their presence plausibly distinguished bills addressing upstream determinants from generic mentalhealth service delivery. Broad, ubiquitously used healthcare terms were deliberately excluded to reduce false positives and preserve domain specificity.

5.4 Analytical Approaches

All analyses treat the introduced bill text as the unit of analysis (one bill = one document) and use text-derived measures to describe and compare patterns of legislative attention rather than to estimate causal effects. Results are therefore interpreted as descriptive, contextual comparisons across states, session years, and party-control categories. Text processing and measurement were implemented primarily in R using the tidytext workflow for tidy token tables and document-term representations (**tidytext?**), with named entity recognition performed via spacyr (wrapping spaCy's en_core_web_lg English model) (**spacyr?**; **spacy?**). Topic modeling was implemented with the topicmodels package using Latent Dirichlet Allocation (LDA) (**topicmodels?**).

To address RQ1 (domain prominence and recurring expressions), the analysis combines dictionary filtering with unigram and n-gram frequency measures. First, dictionary terms and phrases were linked to the token and n-gram tables (unigrams via the token table; bigrams/trigrams via the n-gram table). For RQ1a, domain-term prominence is summarized using term frequency (TF), operationalized as the total count of each dictionary term across the corpus (i.e., summing token counts across bills). For RQ1b, term distinctiveness is summarized with TF–IDF, treating each bill as a separate document so that terms frequent in a small subset of bills are up-weighted relative to terms that occur broadly across the corpus (implemented using tidytext's TF–IDF workflow) (**tidytext?**). For RQ1c, recurring phrases are measured using bigrams and trigrams constructed from bill text, filtered to remove n-grams containing stopwords and formatting artifacts, and then summarized as the most frequent domain-matched n-grams within each domain.

To address RQ2 (variation by state, session year, and party control), the primary outcome is the bill-level domain indicator from the dictionary scoring stage: for each domain, a bill is coded as 1 if it contains at least one dictionary term/phrase and 0 otherwise. Comparisons are reported as shares (proportions) within each subgroup—e.g., the share of bills in a given state-session or party-control category that mention a domain—rather than raw counts, to ensure comparability across groups of unequal size (e.g., New York vs. Ohio; annual vs. biennial session structures). Party control is treated as a contextual grouping variable coded at the opening of each regular session (Democratic trifecta, Republican trifecta, or divided government), and findings are interpreted as descriptive patterns rather than causal party effects.

To address RQ3 (what distinguishes bills with high vs. low social-determinant language), bills were first ranked by the overall domain intensity score (the total number of unique dictionary matches across domains). Bills above the 75th percentile were defined as high-domain, and bills below the 25th percentile were defined as low-domain. Distinctive language was then identified using three complementary approaches. First, unigrams and n-grams were compared across the high- vs. low-domain subsets using class-based TF–IDF, treating the high-domain subset as one aggregated document and the low-domain subset as another, so that terms/phrases characteristic of high-domain bills are up-weighted relative to low-domain bills (**tidytext?**). Second, recurrent bigrams and trigrams were compared using the same contrastive logic to identify short policy-relevant expressions that separate the two groups. Third, named entity recognition (NER) was used to extract organizations and geopolitical entities that function as concrete institutional or place-based signals in high-domain bills (e.g., agencies, programs, state departments). NER was executed with spacyr/spaCy and then cleaned through iterative inspection of the most frequent entities; generic procedural artifacts were removed using a custom entity stoplist. The spaCy LAW entity type was excluded because it predominantly captured internal legislative cross-references and formatting artifacts rather than substantively meaningful named laws, yielding insufficient signal for comparison.

To address RQ4 (latent themes and their relationship to concrete SDOH domains), the analysis fit an LDA topic model to the corpus using a document-term matrix derived from the cleaned token table (**topicmodels?**). Because topic models are sensitive to sparsity and inflectional variation, the topic-model input uses a more normalized representation (lemmatized tokens) than the dictionary and n-gram components. The number of topics was selected as k=20 based on (1) perplexity comparisons across candidate values (k{10,15,20,25,30}) and (2) substantive inspection of the top terms per topic to ensure interpretability. The fitted model produces document–topic proportions () for each bill, which were then joined back to bill metadata and domain scores to examine which topics are most prevalent overall and how topic prevalence co-varies with domain indicators/scores across states and session years. Six topics were identified as procedural/formatting artifacts (state-specific header language, citation conventions, and extraction remnants) and were retained in the topic table for transparency but excluded from substantive RQ4 comparisons.

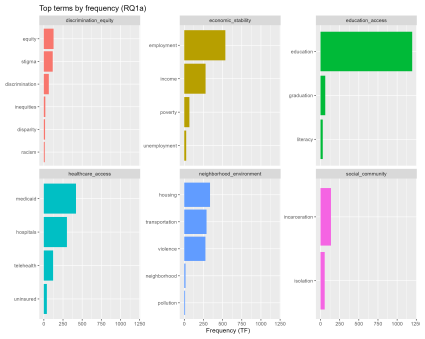

Figure 2: Top domain terms by frequency (RQ1a).

5.5 Validation

Validation combined automated audit checks during corpus assembly with targeted checks for the dictionary, named entities, and topic model. The data pipeline preserved raw API responses, logs, and sample bill texts to confirm that HTML and PDF extraction produced readable content. The SDOH dictionary was reviewed against the literature and the Health People 2030 framework, then refined using a small sample of bills to add missing high-signal terms, remove ambiguous entries, and audit false positives and false negatives in selected domains. NER outputs and LDA topics were validated through inspection of high-frequency entities and top terms, iterative removal of procedural artifacts, and reading bills with the highest topic proportions to confirm that high-proportion topics and dominant entities aligned with substantive bill content rather than formatting noise.

6. Findings

6.1 RQ1

The most frequent dictionary-filtered domain word among introduced bills is education (tf = 1,197), followed by employment (535), medicaid (420), housing (336), hospitals (302), transportation (290), income (277), and violence (276). All of these terms appear in the education access, economic stability, healthcare access, and neighborhood-environment domains. By contrast, terms from the discrimination-equity domain are less frequent, with equity (tf = 127), stigma (tf = 115), and discrimination (tf = 65) as the domain's most frequent terms, while social-community words also have only modest frequency with incarceration (tf = 136) and isolation (tf = 54).

TF-IDF scores highlight a somewhat different pattern: within healthcare access, telehealth (mean TF-IDF = 1.04) and uninsured (0.66) are especially distinctive, and in

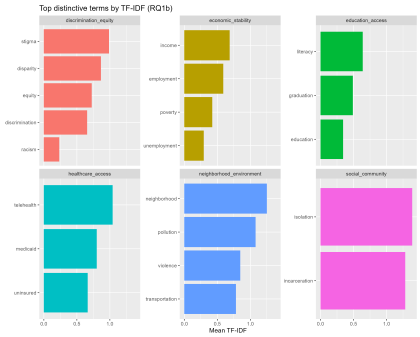

Figure 3: Top domain terms by distinctiveness (TF-IDF, RQ1b).

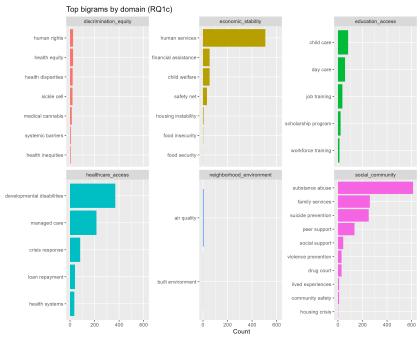

Figure 4: Top domain bigrams in mental health bills (RQ1c).

neighborhood-environment, neighborhood (1.25), pollution (1.08), and violence (0.85) stand out despite not being the most frequent raw tokens. Discrimination-equity terms such as stigma (0.99), disparity (0.86), and equity (0.72), along with social-community terms like isolation (1.39) and incarceration (1.28), also emerge as highly distinctive to the subset of bills that reference them.

Phrase-level patterns reinforce these findings. The most common bigrams are substance abuse (612 occurrences), human services (508), and developmental disabilities (369), which appear in the social-community, economic stability, and healthcare access domains, respectively. The most common trigrams are behavioral health services (241), health care provider (189), and law enforcement agency (180), of which the first two are in the healthcare access domain and the last is in the social-community domain.

6.2 RQ2

State-level comparisons show that, of all states, Washington exhibits the highest shares of bills mentioning most domains, with roughly 72 percent of its mental health bills referencing healthcare access or social-community language

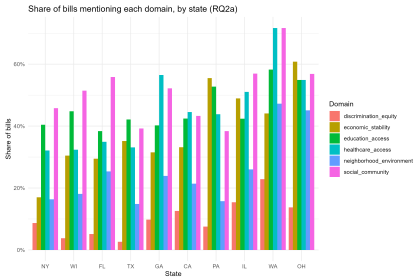

Figure 5: Share of mental health bills mentioning each domain, by state (RQ2a).

and 58 percent mentioning education access, while Ohio stands out for economic stability and neighborhood and built environment language, with about 61 and 45 percent of its bills referencing those domains, respectively.

Across the 2021 and 2023 sessions, domain shares remain relatively stable: education access language rises only slightly from 0.42 to 0.43 of bills, economic stability from 0.328 to 0.332, and healthcare access from 0.40 to 0.42, while social-community references dip modestly from about 0.51 to 0.48. Discrimination-equity and neighborhood-environment mentions show similarly small shifts (roughly 0.11 to 0.09 and 0.22 to 0.21, respectively), suggesting no dramatic reorientation of domain emphasis over the study window.

By party control, divided governments have the highest shares of bills mentioning economic stability (about 45 percent) and education access (about 49 percent), whereas Democratic trifecta states show the highest shares for healthcare access (0.42), neighborhood and built environment (0.22), social-community context (0.51), and discrimination-equity language (0.12). However, Republican trifectas are not far behind on healthcare access (0.38), neighborhood and built environment (0.22), and social-community context (0.58), and they have a slightly higher share of economic stability mentions (0.34) than Democratic trifectas (0.31).

6.3 RQ3

To examine how language differs between bills with higher and lower SDOH intensity, I computed class-based TF-IDF for unigrams and bigrams in the high- and low-domain bill groups. High-domain unigrams skew toward broad policy language, including terms such as equity, sanction, prisoner, and hiv. This pattern suggests that high-domain bills are engaging more directly with problems, sanctions, and conditions, rather than narrow procedural detail. In contrast, low-domain unigrams are more idiosyncratic and proper-noun-like, with examples including welch, emanuel,

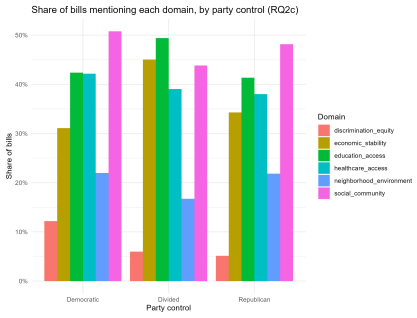

Figure 6: Share of mental health bills mentioning each domain, by party control (RQ2c).

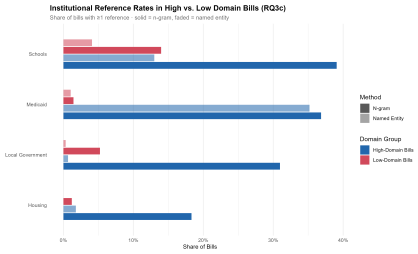

Figure 7: Institutional references in high- vs. low-domain bills (RQ3c).

chris, and tamms, alongside niche medical or procedural terms like hyperbaric, psychosurgical, and mastectomy. This suggests that low-domain bills are more heterogeneous and often anchored in specific projects, locations, or medical technologies rather than a general SDOH framing.

As with unigrams, high-domain bigrams are coherent and policy-relevant, with examples such as certified mental, health assistant, mandatory prison, and designated crisis. Together, these bigrams highlight service delivery and justice or public-safety interfaces around mental health, such as crisis response and prison settings. In contrast, low-domain bigrams are dominated by procedural and fiscal language like appropriations project, project request, and project titled. These bigrams indicate that many low-domain bills are primarily appropriations and project-funding vehicles, with less substantive SDOH language. Overall, high-domain bills show more substantively coherent SDOH-related language in both unigrams and bigrams, whereas low-domain bills mix project- and appropriations-oriented language with idiosyncratic proper nouns and medical terms.

To answer RQ3b, I compared organization (ORG) and geopolitical (GPE) entities between high- and low-domain bills using spaCy NER. I then plotted the top ORGs by total mentions per group, computed high:low mention ratios for ORGs appearing in both groups, and plotted the most frequent GPEs. In the first plot, I found that high-domain bills consistently reference state departments (e.g., Department of Public Health) and public insurance programs (e.g., Medicaid) that directly administer or finance mental health and SDOH-related services, suggesting an institutional focus on service systems and payers. Low-domain bills, by contrast, highlight codes and generic bodies (e.g., The Task Force) rather than specific service agencies or payers, indicating a more procedural emphasis and less direct SDOH content. Surprisingly, the top low-domain "ORG" is a statute reference (Health and Safety Code), not an organization, illustrating NER misclassification. The second plot shows that when these agencies, payers, and program or act names appear at all, they are overwhelmingly concentrated in high-domain bills, with high:low ratios on the order of tens or even hundreds to one. In the GPE plot, both panels are dominated by state and federal references, with only modest differences: high-domain bills reference multiple large states and federal context, while low-domain bills lean slightly more on New York-centered internal references. Overall, high-domain bills disproportionately mention public payers and state health and human-services agencies, whereas low-domain bills instead emphasize statutory codes and task forces, with more NER artifacts, underscoring that they are less centered on SDOH service systems and more on procedural or project-specific matters.

To answer RQ3c, I compared institutional reference rates for four categories—Schools, Medicaid, Local Government, and Housing—in high- vs. low-domain bills using both n-gram pattern matching and entity-based matching (spaCy ORG/GPE; see Figure 7). Both methods agree that Medicaid references are dramatically more common in high-domain bills (roughly an order of magnitude) and that schools are mentioned about two to three times as often in high-domain bills as in low-domain bills. N-grams also show a clear gap for local government and housing, with high-domain bills much more likely to reference these institutions, while entity-based detection is sparse for these categories, suggesting local-government and housing references are often embedded in phrases that NER does not tag cleanly. Across all four categories, high-domain bills have higher institutional reference rates than low-domain bills, most strikingly for Medicaid and schools.

6.4 RQ4

To answer RQ4a, I used an LDA model to estimate document–topic proportions and examined both mean topic prevalence and dominant-topic counts. The most prevalent topics are Education & Community, Advisory Bodies & Task Forces, Insurance & Medicaid Coverage, Criminal Justice,

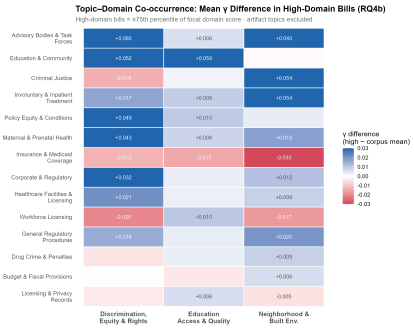

Figure 8: Topic Domain Heatmap

and Workforce Licensing, and together these cover roughly two thirds of bills as their dominant topics. The distribution is highly concentrated rather than flat, and explicitly clinical-care or facilitylicensing topics sit near the bottom of the prevalence rankings, suggesting that contemporary state mental health legislation in this corpus is oriented more toward education settings, advisory structures, coverage and financing, justice-system interfaces, and workforce regulation than toward narrow clinical or facility issues.

To answer RQ4b, I compared topic proportions with SDOH domain scores and produced a topic–domain heatmap of alignment (mean document–topic proportion differences between high-domain bills and the full corpus; see Figure 8). For Discrimination, Equity & Rights, the strongest alignment is with Advisory Bodies & Task Forces, with secondary alignment in Education & Community and Policy Equity & Conditions, indicating that equity concerns are most often routed through commissions and oversight mechanisms rather than a dedicated equity topic. For Education Access & Quality, the Education & Community topic shows the strongest positive alignment, providing a face-validity check that education-domain language concentrates in the education-themed topic. For Neighborhood & Built Environment, the strongest alignments are with Criminal Justice and Involuntary & Inpatient Treatment, with secondary alignment in Advisory Bodies & Task Forces, suggesting that neighborhood concerns are frequently intertwined with justice-system procedures, involuntary treatment provisions, and advisory structures. Cross-domain, Advisory Bodies & Task Forces and Involuntary & Inpatient Treatment show positive alignment across multiple domains, while Insurance & Medicaid Coverage shows negative alignment for all three focal domains, meaning that bills most strongly framed around education, equity, or neighborhood SDOH tend to devote less text to insurance and payer language than the corpus average.

To address RQ4c, I treated the state and year analyses as exploratory context rather than primary tests. The year trend plot shows that all four SDOH-focal topics—Education & Community, Criminal Justice, Insurance & Medicaid Coverage, and Policy Equity & Conditions—are present in each session year and change only gradually over time. Mean topic proportions increase slightly from 2021 to 2023 for Education & Community and for the other focal topics, but there is no sharp break or reversal, suggesting incremental rather than dramatic growth in SDOH framing over this short window.

At the state level, the heatmaps and stateŒdomain interaction plot show that several topics are represented in every state but with modest differences in emphasis. Education & Community and Insurance & Medicaid Coverage are relatively prominent in larger states such as California, Illinois, and New York, while Involuntary & Inpatient Treatment and Criminal Justice are more concentrated in a subset of states, including Ohio, Washington, and Georgia. Within states, mean domain scores and aligned-topic proportions generally move in the same direction but not in lockstep, indicating that some states have stronger topic–domain alignment for particular SDOH domains than others. Given the small number of years and the limited set of states in this corpus, I interpret RQ4c descriptively: SDOH-aligned topics appear across states and years, with moderate variation in emphasis, but the data do not show a clear, uniform temporal trend or a single "leader" state in topic–domain alignment.

7. Discussion

7.1 Interpreting SDOH domain attention (RQ1 + RQ2)

Across RQ1 and RQ2, the results show that SDOH language in state mental health bills is unevenly distributed across domains, but that each domain has both highfrequency and more specialized terms. In RQ1a and RQ1c, the most frequent unigrams and bigrams tend to come from education access, economic stability, health care access, neighborhood and built environment, and social and community domains—for example, education, employment, Medicaid, housing, transportation, substance abuse, and human services all appear often—but each domain also contains less common terms such as literacy, neighborhood, and health equity that occur relatively rarely. RQ1b shows that several of these lowerfrequency terms, including literacy in education access, neighborhood and pollution in neighborhoodenvironment, and stigma and disparity in the discriminationequity domain, have high mean TFIDF scores, indicating that they are distinctive when they appear and tend to cluster in par-

ticular bills rather than diffusing across the corpus. Taken together, these patterns suggest that while all six domains are represented and share a mix of common and specialized language, some aspects of SDOH—especially certain discriminationequity and neighborhood concepts—enter mental health legislation through more targeted, billspecific vocabulary rather than broad, ubiquitous terms. paste.txt

RQ2 adds that domain attention also varies systematically by state and party control. Across states and partycontrol types, discrimination, equity, and rights and neighborhood and built environment consistently appear among the lowestshare domains, whereas education access and quality, health care access and quality, and social and community context are more routinely mentioned, even though their specific terms and phrases differ by context. These descriptive patterns indicate that both state context and partisan control align with different emphases within a shared SDOH framework, but neither factor alone fully explains crossstate variation in which domains are foregrounded in mental health bills.

7.2 High- vs. low-domain bills (RQ3)

The high- versus low-domain contrast in RQ3 shows that SDOH intensity is tied to different kinds of distinctive bill language: high-domain bills pair SDOH language with justice, crisis-response, and service-system terms, whereas low-domain bills concentrate on project scaffolding and narrow technical or locality-specific details, suggesting that low-domain bills invoke SDOH language more superficially and are less centered on upstream social conditions than high-domain bills. These lexical differences are mirrored in the institutional references that appear most often in each group. High-domain bills invoke public payers and state health and human-services agencies far more often than low-domain bills, which instead highlight generic task forces and statutory code labels. Geopolitical entities, in contrast, show few differences between high- and low-domain bills, suggesting that place-based language is common across the corpus and does not meaningfully differentiate SDOH intensity. Finally, the institutional reference comparison for specific categories (Schools, Medicaid, Local Government, Housing) shows that high-domain bills are much more likely to reference schools, Medicaid, local government, and housing, underscoring that SDOH-intensive legislation is tightly connected to key service and governance actors.

7.3 Topics and domains together (RQ4)

The topic model results in RQ4 indicate that most introduced state mental health bills cluster around a small set of institutional and systems-oriented themes rather than being evenly distributed across many topics. Linking topics to SDOH domain scores shows that education access and quality aligns cleanly with the Education & Community topic, discrimination, equity, and rights aligns most strongly with Advisory Bodies & Task Forces and Policy Equity & Conditions, and neighborhood and built environment aligns with Criminal Justice and Involuntary & Inpatient Treatment, with advisory bodies again playing a supporting role. Advisory Bodies & Task Forces and Involuntary & Inpatient Treatment therefore play cross-domain roles, appearing more often in bills with high education, neighborhood, and equity scores, while the Insurance & Medicaid Coverage topic shows negative alignment with all three focal domains, indicating that bills framed most strongly around education, equity, or neighborhood SDOH tend to devote less text to insurance and payer language than the corpus average.

Taken together, the domain and topic results suggest that state mental health bills most often frame upstream determinants through education, health care access, and social and community context, while discrimination, equity, and neighborhood conditions appear less frequently and more concentrically in specific bills. Legislatures and advocates aiming to advance mental health equity could use these patterns to press for more routine incorporation of equity and neighborhood language into mainstream mental health legislation and to tie upstream SDOH framing more explicitly to payer, coverage, and reimbursement provisions so that equity, education, and neighborhood concerns are reflected in the mechanisms that finance and deliver services. Equity-related language's strong alignment with Advisory Bodies & Task Forces and Policy Equity & Conditions indicates that commissions and task forces are common vehicles for addressing discrimination, equity, and rights, but the findings also point to the importance of pairing advisory mechanisms with statutory changes to programs, funding, and enforcement if equity considerations are to move beyond consultation into implementation.

8. Limitations

This project has several limitations that shape how its findings should be interpreted. First, the SDOH domain dictionary is a necessary simplification of a much richer conceptual space; even with literature-based term selection, inductive corrections, and targeted validation checks, some relevant language is missed and some borderline phrases are misclassified, especially in domains such as discrimination and neighborhood conditions. Second, the corpus itself is restricted to ten states, introduced bill versions, and regular sessions between 2021 and 2024, meaning that the domain

patterns observed here may not generalize to other jurisdictions, bill types, or time periods. Third, offtheshelf models for named entities and topic modeling introduce artifacts: spaCy mislabels some statutes and formatting as organizations and places, and several LDA topics capture procedural or formatting conventions rather than substantive themes despite efforts to flag and exclude them from key comparisons. Finally, the comparative analyses by state, session year, and party control are descriptive rather than causal, so observed differences should be read as patterns of legislative attention, not as evidence that partisan control or state context directly causes changes in SDOH language.

9. Future Research

Future research could extend this analysis by expanding the corpus to additional states, longer time windows, and other types of mental-health-adjacent legislation to test whether these domain and topic patterns generalize beyond the current ten-state sample. Methodologically, richer SDOH measures and improved institutional recognition—such as combining dictionary approaches with supervised or embedding-based models and domain-specific NER—could improve coverage of discrimination, equity, and neighborhood language while reducing artifacts. Linking legislative attention patterns to implementation or outcome data would also help future work move beyond mapping how SDOH appear in bill text toward understanding whether and how those textual patterns relate to mental health policy change.

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